

Notes on Variance Calculations

The mathematical derivations showing how we can achieve the following in our panel data simulations:

- The variance of X , Y and C is 1.
- Each variable has a mean 0.
- The autoregressive effects of X_{t-1} on X_t is 0.25 (same for Y_{t-1} on Y_t).
- The lag-1 cross effect of X_{t-1} on Y_t is 0.10 (same for Y_{t-1} on X_t).
- There are 3 confounders, and hence $k = 3$.
- Their effects differ for X and Y , such that X is affected at $[0.30, 0.20, 0.10]$ and Y is affected at $[0.30, 0.20, 0.10]$.
- The innovations have residual correlations of 0.3 at each time-point, but not across time-points.
- At the first time-point ($T = 0$), X and Y are correlated due to the confounding variables.
- At the first time-point, we can assume that the dynamic process was going on before the data started being collected. Hence, the 'observed' covariance matrix S between X and Y should be the same for the first measurement occasion. But since we do not have any previous time-point creating feed forward effects, X and Y should be simulated with 'extra' covariance as if they were 'caused' by previous variables.
- One possible extension: confounders in reality are with a very high probability not uncorrelated. In the event that these are correlated, we get a true matrix Ψ instead of the identity matrix I . Calculations are not affected with respect to unknown values.

What we need to do, is to find the right variance for the innovations such that the total variance of X and Y becomes 1. We also need to find out what correlation X and Y are supposed to have at each time-point. As such we have three unknowns: a variance of the innovation for X , a variance of the innovation for Y , and a correlation between X and Y . We must pick these values such that all our demands are met. In essence, this is the purpose of all math done here.

Defining matrices

We have variables X over occasions T and Y over occasions T . This yields the vector:

$$Z_t = \begin{bmatrix} X_t \\ Y_t \end{bmatrix}$$

For one individual i , the vector looks like:

$$Z_i = \begin{bmatrix} X_{i1} \\ Y_{i1} \\ X_{i2} \\ Y_{i2} \\ X_{i3} \\ Y_{i3} \\ X_{i4} \\ Y_{i4} \\ X_{i5} \\ Y_{i5} \end{bmatrix}.$$

Expanding for N individuals:

$$Z = \begin{bmatrix} X_{1,1} & Y_{1,1} & X_{1,2} & Y_{1,2} & X_{1,3} & Y_{1,3} & X_{1,4} & Y_{1,4} & X_{1,5} & Y_{1,5} \\ X_{2,1} & Y_{2,1} & X_{2,2} & Y_{2,2} & X_{2,3} & Y_{2,3} & X_{2,4} & Y_{2,4} & X_{2,5} & Y_{2,5} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ X_{N,1} & Y_{N,1} & X_{N,2} & Y_{N,2} & X_{N,3} & Y_{N,3} & X_{N,4} & Y_{N,4} & X_{N,5} & Y_{N,5} \end{bmatrix}.$$

Now for time-point t , X_t and Y_t are made up of some part of the confounder effect, and some part by the dynamic effects. With the dynamic effects here we mean every effect that 'feeds forward' to the next time-point. For the first time-point, there are no dynamic effects at all, since there are no previous time-points. Since we assume the process does not start with measurement, the observed correlation matrix between X and Y should be unaffected though. Another way of phrasing this is that some part of the variance in X (or Y) is explained by the confounders, some part is explained by the dynamic effects, and some part is explained by the innovations. We can capture this in the formula:

$$Z_t = AZ_{t-1} + BU + E_t,$$

where

- A is the matrix of autoregressive and cross-lagged effects,
- B is the matrix of confounder effects,
- U is the vector of time-invariant confounders,

- E_t is the vector of innovations at time-point t .

Here we can clearly see that if there is no previous time-point, the dynamic effects part (AZ_{t-1}) will be zero. If Z_t is then to have a similar covariance matrix, the part of variance received from the previous time-points and the lagged effects should be 'compensated'.

If we choose our effects as denoted above we get the following A matrix:

$$A = \begin{bmatrix} a & b \\ b & a \end{bmatrix} = \begin{bmatrix} 0.25 & 0.10 \\ 0.10 & 0.25 \end{bmatrix}$$

We get the following beta matrix:

$$B = \begin{bmatrix} c_{x1} & c_{x2} & c_{x3} \\ c_{y1} & c_{y2} & c_{y3} \end{bmatrix} = \begin{bmatrix} 0.30 & 0.20 & 0.10 \\ 0.30 & 0.20 & 0.10 \end{bmatrix}$$

And matrix U :

$$U = \begin{bmatrix} c_{1,1} & c_{1,2} & c_{1,3} \\ c_{2,1} & c_{2,2} & c_{2,3} \\ c_{3,1} & c_{3,2} & c_{3,3} \\ \vdots & \vdots & \vdots \\ c_{N,1} & c_{N,2} & c_{N,3} \end{bmatrix}$$

Finally, we can specify the innovations matrix as:

$$E_t = \begin{bmatrix} e_{X_t} \\ e_{Y_t} \end{bmatrix}$$

Confounder variance calculations

As we just denoted, each measurement occasion takes some variance from the confounders and some from the dynamic effect. The variance received from the confounders is time-invariant, which means we can calculate it directly. Since the variances of all variables (and thus also the confounders) is 1, and the confounders are not correlated with each other, their covariance matrix is the identity matrix I . Note that including confounder covariance does not change very much about the calculations, since it is a value we just pick and insert using Ψ . Hence, the variance received from the confounders is:

$$S_U = B\Psi B^T = B I B^T = B B^T$$

If we use the values from above, we get:

$$S_U = \begin{bmatrix} 0.30 & 0.20 & 0.10 \\ 0.30 & 0.20 & 0.10 \end{bmatrix} \begin{bmatrix} 0.30 & 0.30 \\ 0.20 & 0.20 \\ 0.10 & 0.10 \end{bmatrix} = \begin{bmatrix} 0.14 & 0.14 \\ 0.14 & 0.14 \end{bmatrix}$$

The innovation covariance matrix

The confounders add a variance of 0.14 to both X and Y, and a covariance of 0.14 between X and Y at each time-point. We wanted each variable to have a variance of 1, which means that the dynamic effect still needs to add $1 - 0.14 = 0.86$ to the variance of each variable. We also want the innovations to have a correlation of 0.3 at each time-point, which means that the part of X and Y that cannot be explained by X_{t-1} and Y_{t-1} or C needs to be correlated at 0.3. In math notation, we want:

$$\text{Corr}(e_{X_t}, e_{Y_t}) = \rho = 0.3$$

We can denote the variance of the innovations as:

$$\text{Var}(e_{X_t}) = v_1 \quad \text{and} \quad \text{Var}(e_{Y_t}) = v_2$$

The covariance between the innovations is then their correlation multiplied by their standard deviations:

$$\text{Cov}(e_{X_t}, e_{Y_t}) = \rho\sqrt{v_1}\sqrt{v_2}$$

The covariance matrix of the innovations can then be written as:

$$S_E = \begin{bmatrix} v_1 & \rho\sqrt{v_1v_2} \\ \rho\sqrt{v_1v_2} & v_2 \end{bmatrix}$$

Since we know that $\rho = 0.3$, we can rewrite this as:

$$S_E = \begin{bmatrix} v_1 & 0.3\sqrt{v_1v_2} \\ 0.3\sqrt{v_1v_2} & v_2 \end{bmatrix}$$

Which leaves us with two unknown values: v_1 and v_2 . We need to choose these such that the total variance of X and Y becomes 1. Another way of phrasing this requirement is that S , which is the total variance-covariance matrix of Z_t , needs to have 1s on the diagonal.

The covariance matrix S (of X and Y)

Now first we need to specify what S looks like. We know that:

$$S = \text{Var}(Z_t) = \begin{bmatrix} \text{Var}(X_t) & \text{Cov}(X_t, Y_t) \\ \text{Cov}(X_t, Y_t) & \text{Var}(Y_t) \end{bmatrix}$$

We want the variance of X and Y to be 1, which means that we want:

$$\text{Var}(X_t) = \text{Var}(Y_t) = 1$$

yielding:

$$S = \begin{bmatrix} 1 & \text{Cov}(X_t, Y_t) \\ \text{Cov}(X_t, Y_t) & 1 \end{bmatrix}$$

We can now denote $\text{Cov}(X_t, Y_t)$ as c . This yields:

$$S = \begin{bmatrix} 1 & c \\ c & 1 \end{bmatrix}$$

Note that this covariance matrix is stationary when it is the same for all time-points. In order to satisfy this constraint, the following equation must hold:

$$S = ASA^T + S_U + S_E$$

The covariance matrix of the current time-point is equal to that of the previous time-point, multiplied by the cross-lagged and autoregressive effects, plus the variance received from the confounders, plus the variance received from the innovations. We must solve this equation for v_1 , v_2 and c . v_1 and v_2 are only present in S_E , while c is only present in S . Expanding the equation yields:

$$\begin{bmatrix} 1 & c \\ c & 1 \end{bmatrix} = \begin{bmatrix} (ASA^T)_{11} & (ASA^T)_{12} \\ (ASA^T)_{21} & (ASA^T)_{22} \end{bmatrix} + \begin{bmatrix} 0.14 & 0.14 \\ 0.14 & 0.14 \end{bmatrix} + \begin{bmatrix} v_1 & \rho\sqrt{v_1v_2} \\ \rho\sqrt{v_1v_2} & v_2 \end{bmatrix}.$$

Note that on the left side is our desired stationary variance-covariance matrix between X and Y , containing unknown c . On the right side we have the part carried over from the previous wave, the part contributed by the confounders, and the part added by the innovations. We can rewrite this as a system of equations:

$$1 = (ASA^T)_{11} + 0.14 + v_1$$

$$1 = (ASA^T)_{22} + 0.14 + v_2$$

$$c = (ASA^T)_{12} + 0.14 + \rho\sqrt{v_1v_2}$$

Note that the two off-diagonal elements are equal, since covariance matrices are symmetric. As such, the equation for c only yields one equation. We should now compute $ASA^\top$ for all these equations. Recall that A is:

$$A = \begin{bmatrix} a & b \\ b & a \end{bmatrix} = \begin{bmatrix} 0.25 & 0.10 \\ 0.10 & 0.25 \end{bmatrix}$$

Now given the positions of a and b in the matrix, the equations simplify to:

$$(ASA^\top)_{11} = a^2S_{11} + 2abS_{12} + b^2S_{22}$$

$$(ASA^\top)_{22} = a^2S_{11} + 2abS_{12} + b^2S_{22}$$

$$(ASA^\top)_{12} = abS_{11} + (a^2 + b^2)S_{12} + abS_{22}$$

Now note that $S_{11} = S_{22} = 1$ and $S_{12} = c$. This yields:

$$(ASA^\top)_{11} = a^2 + 2abc + b^2$$

$$(ASA^\top)_{22} = a^2 + 2abc + b^2$$

$$(ASA^\top)_{12} = 2ab + (a^2 + b^2)c$$

Since we know $a = 0.25$ and $b = 0.10$, we can insert these values:

$$(ASA^\top)_{11} = 0.25^2 + 2(0.25)(0.10)c + 0.10^2 = 0.0725 + 0.05c$$

$$(ASA^\top)_{22} = 0.0725 + 0.05c$$

$$(ASA^\top)_{12} = 2(0.25)(0.10) + (0.25^2 + 0.10^2)c = 0.05 + 0.0725c$$

We can plug these into our system of equations:

$$1 = (0.0725 + 0.05c) + 0.14 + v_1$$

$$1 = (0.0725 + 0.05c) + 0.14 + v_2$$

$$c = (0.05 + 0.0725c) + 0.14 + \rho\sqrt{v_1v_2}$$

Now note that the first two equations are only different in v_1 and v_2 . The left side of the equations is the same. From these two facts we can conclude that $v_1 = v_2$. The first two equations are thus the same. We can now solve for v and rewrite this single equation as:

$$v = 1 - 0.14 - 0.0725 - 0.05c = 0.7875 - 0.05c$$

The variances of the innovations are thus equal and given by $v = 0.7875 - 0.05c$. We can now plug this formula for v into the third equation. First we use that $v_1 = v_2 = v$, yielding:

$$c = (0.05 + 0.0725c) + 0.14 + \rho v$$

Since $\rho = 0.3$, we can rewrite this as:

$$c = 0.19 + 0.0725c + 0.3v$$

Now we can plug in the formula for v :

$$c = 0.19 + 0.0725c + 0.3(0.7875 - 0.05c)$$

Solving this yields:

$$c = \frac{0.19 + 0.7875 \cdot 0.3}{0.9275 + 0.05 \cdot 0.3} = 0.4523$$

This result can now be plugged back into the formula for v :

$$v = 0.7875 - 0.05 \cdot 0.4523 = 0.7644$$

And now we can populate the matrix S_E :

$$S_E = \begin{bmatrix} v & \rho v \\ \rho v & v \end{bmatrix} = \begin{bmatrix} 0.7644 & 0.2295 \\ 0.2295 & 0.7644 \end{bmatrix}$$

And we can plug the result for c back into the matrix S :

$$S = \begin{bmatrix} 1 & c \\ c & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0.4523 \\ 0.4523 & 1 \end{bmatrix}$$

And now all our matrices are fully known, and we can simulate data!